

Agile Methodologies

Lecture 04

1

Agile Methodologies

- There's a big gap between understanding the values and principles in the Agile Manifesto and actually changing the way your team builds software.
- There are agile **methodologies** that are actually intended to help teams adopt agile and improve their projects.

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2

2

Agile Methodologies

- Agile methodologies are valuable to you because they help you see the practices in context.
- They are especially useful for teams that are not familiar with all of the agile practices.



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3

3

Agile Methodologies

- Each methodology was developed and improved over the years by agile expert.
- And these are focused on all of the different parts of the elephant.



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4

Agile Methodologies

- Adopting a methodology in full means project team will be following a tried-and-true path.
- This path that takes project team from the beginning to the end of a software project, without the trial and error that can lead to a fractured perspective.



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5

5

Agile Methodology

- An agile methodology is a collection of practices combined with ideas, advice, and often a body of knowledge and wealth of experience among agile practitioners.



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6

Agile Methodology

- An agile methodology will outline different roles and responsibilities for everyone on the project.
- And it will recommend certain practices for each of them at various stages of the project.

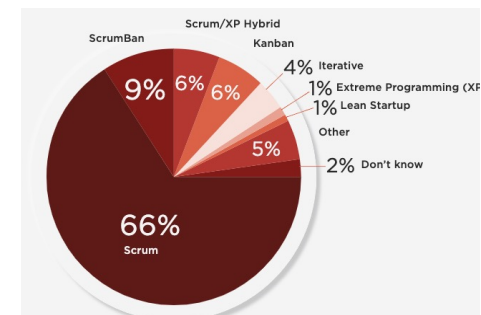


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7

7

Popular Agile Methodologies

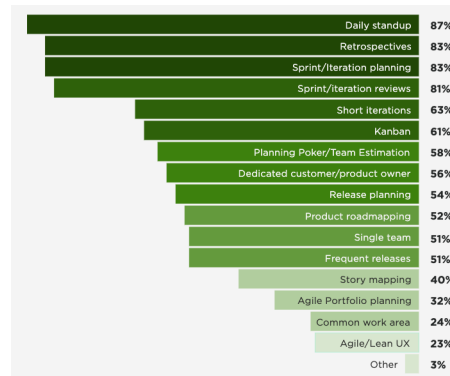
Source: 15th State of Agile Report, 2021

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Popular Agile Practices

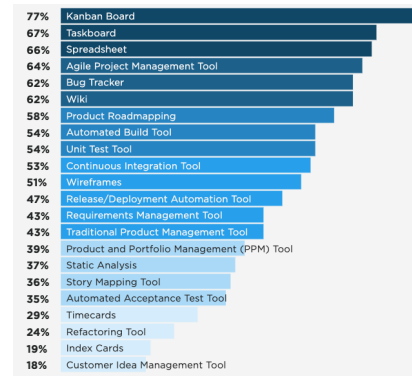


Source: 15th State of Agile Report, 2021

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Popular Agile Tools

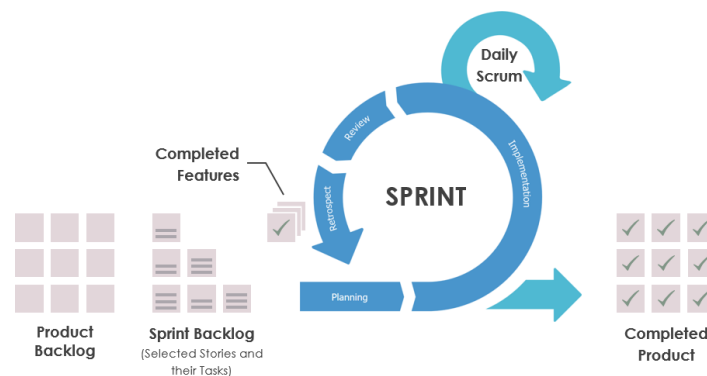


Source: 15th State of Agile Report, 2021

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SCRUM



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11

11

eXtreme Programming (XP)

- Using simultaneous phases, an XP team produces deployable software every week.
- In each iteration, the team analyzes, designs, codes, tests, and deploys a subset of features.
- Many XP teams use iterations that last one week, but others use two-week or month-long iterations.

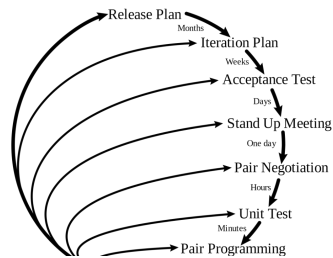
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12

12

eXtreme Programming (XP)

- XP prescribes specific development practices aimed at improving collaborating with users, planning, developing, and testing.



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13

SCRUM and XP

- Scrum and XP have many things in common.
- Most common one is both being iterative.
- Length of iterations can vary from a week to a month period.
- Putting a limit on how long an iteration lasts is called **timeboxing**.
- It helps users to know when they can expect new features to be delivered.

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14

14

What is Lean ?

- Lean is not a methodology.
- Rather, it's a name for a mindset.
- It has its own set of values, and thinking tools to help you adopt it.



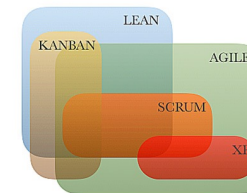
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15

15

Kanban

- Kanban is an agile methodology for improving the way that a team builds software.
- It's built on the values of Lean.
- It includes its own practices to help a team improve and evolve.



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16

16

Scrumban

- Scrumban combines important features of two popular agile methodologies: Scrum and Kanban.
- It merges the structure and predictability of Scrum with Kanban's flexibility and continuous workflow.
- It can help a team benefit from both the prescriptive nature of Scrum and the freedom of Kanban to improve their processes.

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17

17

Focus of Agile Methodologies

- Different Agile methodologies have different practices and focus.
- How is it possible that these different approaches to agile can have entirely different focus and practices, yet all be considered agile?

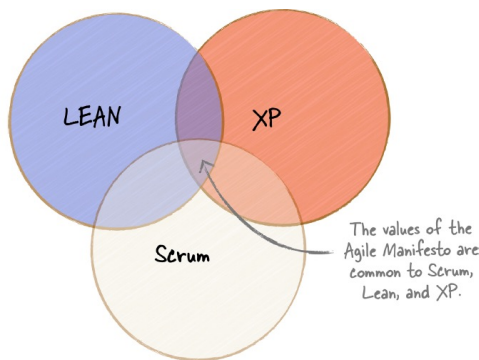


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18

18

Focus of Agile Methodologies



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19

19

Agile Methodologies or Practices

- Many teams find adopting methodologies to be more effective than simply adopting individual practices.
- While adopting individual practices allows each team member to select those practices specific to his or her job, adopting an entire methodology encourages the entire team to get together and figure out as a team how to adopt all of the practices for that methodology.

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20

20

Agile Methodologies or Practices

- To do this, they need to change the way they think about doing their jobs.
- Methodologies are constructed around the agile values and principles, so this change in attitude is generally toward collaboration and interactions, working software, and responding to change.